

Astrodynamics I: Midterm Exam

April 17, 2015 1:05-2:35PM (90 minutes)

Closed Books/Notes

1. (30) Three-Body and N -Body Equations of Motion

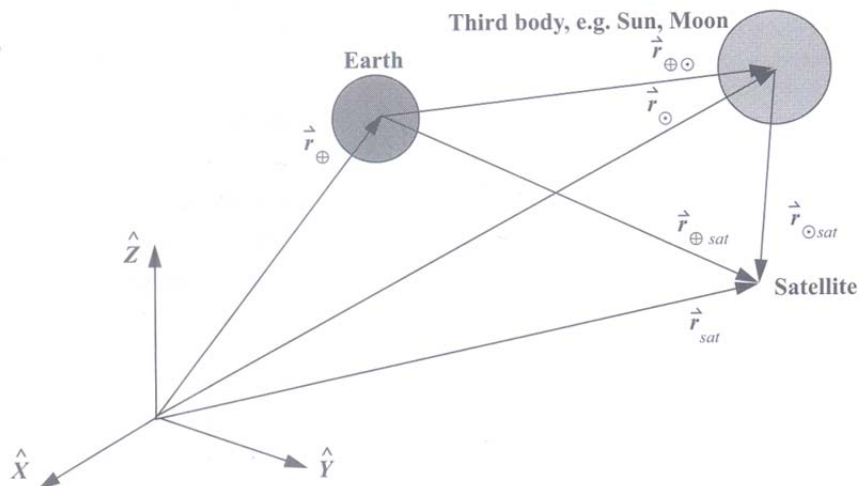


Figure 1-15. Three-body Geometry. The inertial coordinate system XYZ is used to measure all the accelerations. The double subscripts are necessary to completely define the direction of each vector. The satellite is intended to be orbiting the Earth, but for clarity, I've greatly exaggerated the figure.

- (10) Refer to the notations and descriptions in Figure 1-15 to derive the three-body equations of motion in terms of inertial (not relative) acceleration of each body.
- (10) Use the index notation to extend the three-body equations of motion into the general n -body equations of motion.
- (10) Mathematically state the ten known integrals of general n -body problem succinctly and precisely without proof.

2. (40) Kepler's Equation

- (10) Refer to the notations and their geometric relations in Figure 2-2 to derive $\sin \nu$ and $\cos \nu$ as functions of eccentric anomaly E and eccentricity e .

Hint: You may use $BC = B'C\sqrt{1-e^2}$ without proof.

- (20) Start with the conservation of magnitude of specific angular momentum to derive Kepler's equation in basic form:

$$t - T = \sqrt{\frac{a^3}{\mu}} (E - e \sin E)$$

- c. (10) Introduce the notion of mean motion $n \equiv \sqrt{\mu / a^3}$ and mean motion $M \equiv n(t - T)$ to rewrite Kepler's equation in general form which accounts for (1) the spacecraft **NOT** necessarily located at periapsis at initial time and (2) multiple revolutions.

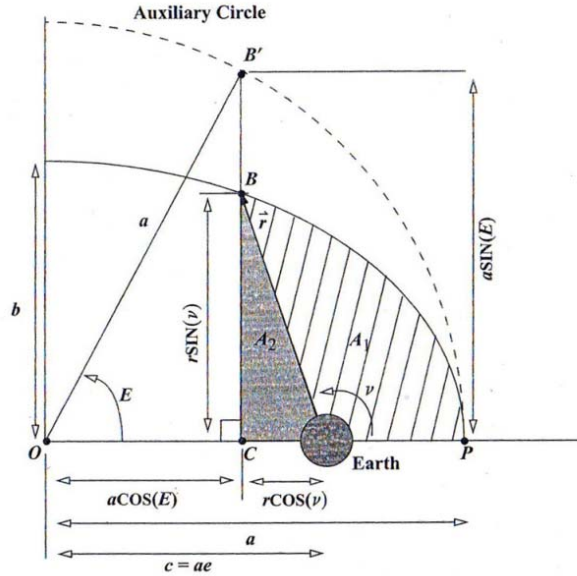


Figure 2-2. **Geometry of Kepler's Equation.** The eccentric anomaly uses an auxiliary circle as shown. The ultimate goal is to determine the area, A_1 , which allows us to calculate the time.

3. (10) State the definition of the following terminologies **succinctly** and **precisely**.
- (5) Argument of Latitude (u) for Circular Inclined Orbit
 - (5) True Longitude (λ_{true}) for Circular Equatorial Orbit

4. (20) **Conversion between Classical Orbital Elements and State Vectors**

Clearly explain the procedure of how to obtain the modified classical orbital elements $(a, e, i, \tilde{\omega}_{true}, \nu)$ from the state vectors $(\vec{r}, \vec{v})$ for the elliptical equatorial orbit.

Hint: You may use $\vec{e} = \frac{1}{\mu} \left[\left(v^2 - \frac{\mu}{r} \right) \vec{r} - (\vec{r} \cdot \vec{v}) \vec{v} \right]$ and/or $e = \sqrt{1 + \frac{2Eh^2}{\mu_E^2}}$ without proof.